

ASD Data Analytics: Feature Engineering, Visualizations, and Statistical Framework

This document details the comprehensive engineering framework, visualization paradigms, and statistical validation utilized in the *Beyond the Diagnosis* ASD analytics project.

1. Composite Severity Scores

These scores shift the analytical paradigm from a restrictive binary model to a continuous spectrum, capturing the density and graduation of atypical presentations.

Feature Name	Calculation / Logic	Clinical Purpose	Visualization & Interpretation
Total Atypical Trait Burden	Summation of all behavioral questions (A1 through A10), scaling from 0 to 10.	Quantifies the absolute density of atypical presentations across the core spectrum.	Bimodal Histogram: Diagnosed individuals heavily cluster above a score of 6, while undiagnosed individuals cluster below 4. This establishes a clear structural diagnostic threshold.
Social Interaction Score	Aggregates responsive attention, spontaneous eye contact, and joint attention features (A1 + A2 + A6). Scale: 0 to 3.	Isolates and measures foundational continuous social capacity. Higher scores denote superior social orientation.	Grouped Bar Chart: Compares average social scores across age groups and gender to identify precise gaps in developmental social engagement.
Communication Deficit Score	Applies inverse coding to request-pointing, sharing-pointing, and common gesture usage: (1-A3) + (1-A4) + (1-A9). Scale: 0 to 3.	Measures non-verbal communication failures. Flipped coding transforms the absence of a milestone into the measurable presence of a clinical deficit.	Box Plot: Displays median deficit scores and operational outliers broken down by diagnostic status, highlighting starkly polarized communicative divides.

Feature Name	Calculation / Logic	Clinical Purpose	Visualization & Interpretation
Behavioral Atypicality Score	Combines inverse-coded normative features (imagination, empathy, typical communication style) with a direct atypical marker (space-staring): $(1-A5) + (1-A7) + (1-A8) + A10$. Scale: 0 to 4.	Synthesizes restricted, repetitive, or internally focused behavioral patterns.	Heatmap: Cross-references behavioral scores against diagnosis cross-tabulated by gender, exposing subtle gender-based behavioral variations.

2. Clinical Profiles & Archetypes

These multi-dimensional flags evaluate complex presentation dynamics, contextual masking, and phenotypic classifications.

Feature Name	Calculation / Logic	Clinical Purpose	Visualization & Interpretation
Social-Comm Imbalance Profile	Evaluates the directional gap between structural social motivation and functional non-verbal communication tools.	Differentiates individuals who desire interaction but lack tools ("Socially Motivated but Comm Blocked") from those possessing basic mechanics but lacking social impetus ("Capable but Withdrawn").	Stacked Bar Chart: Illustrates the relative proportion of these clinical profiles within each diagnostic category.
Eye Contact Discordance	Flags logical contradictions between spontaneous eye contact and responsive attention orienting.	Uncovers context-dependent masking behaviors or forced social mimicry patterns.	Clustered Bar Chart: Maps out "Fully Typical", "Global Deficit", and "Discordant/Masked" cohorts among diagnosed males and females to isolate gender skew.

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Empathy–Imagination Profile	Contrasts measures of emotional responsiveness against indicators of cognitive flexibility and pretend play.	Maps specific internal processing impairments by delineating isolated emotional deficits from global cognitive-emotional delays.	Grouped Column Chart: Track the distribution of these processing variations across different generational and developmental age groups.
Behavioral Archetype	Master clustering feature that groups individuals into high-level phenotypes (e.g., Severe Global Delay, Communication Impaired, Socially Withdrawn, Mild/Typical).	Synthesizes complex clinical features into an actionable, unified phenotype for executive reporting and clinical triage workflows.	100% Stacked Bar: Demonstrates the precise empirical diagnosis rate within each distinct behavioral archetype cluster.

3. Healthcare Proxy Flags (Audit Metrics)

Binary tracking metrics designed to discover systemic screening gaps, hidden masking behaviors, and clinical false negatives.

Feature Name	Calculation / Logic	Clinical Purpose	Visualization & Interpretation
Late Diagnosis Flag	Triggers true if Age $\geq$ 18 AND the screening yields a Positive ASD Diagnosis.	Identifies adults who navigated their development without pediatric detection—serving as a metric for systemic screening failure.	Bar Chart: Compares late-diagnosed versus early-diagnosed rates by gender to show which demographics are routinely missed in youth.
Masking Proxy Flag	Triggers true if an individual has a Positive ASD Diagnosis AND explicitly self-reports a "Typical"	Detects active behavioral camouflaging or a profound lack of personal deficit awareness.	Pie Chart: Quantifies the absolute scale of the masking phenomenon by visualizing the proportion of

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	communication style.		diagnosed individuals reporting typical communication.
Sub-Clinical Missed Flag	Triggers true if Total Atypical Trait Burden ≥ 7 AND the clinical outcome is listed as a Negative Diagnosis.	Serves as a rigorous data audit to intercept critical diagnostic false negatives—individuals possessing massive behavioral loads who went undetected.	Scatter Plot: Highlights these specific cases as critical outlying points located far above the clinical severity threshold within the negative-diagnosis sub-population.
Compensatory Masking Flag	Triggers true if Age ≥ 13 , Total Atypical Trait Burden ≥ 5 , AND the individual maintains a perfect structural Social Score.	Isolates individuals who have consciously or unconsciously learned to force neurotypical social milestones, masking severe underlying trait burdens.	Grouped Chart: Cross-examines rates by age and gender, frequently revealing a starkly female-skewed clinical distribution.

4. Risk, Genetics & Diagnostic Proximity

Integrates biological background, hereditary foundations, and proximity to clinical boundaries.

Feature Name	Calculation / Logic	Clinical Purpose	Visualization & Interpretation
Genetic Severity Matrix	Fuses documented family history of ASD with recorded behavioral trait burdens.	Evaluates if high-severity clinical profiles are strongly correlated with hereditary lines or present spontaneously.	Heatmap: Maps genetic penetrance bands (High, Intermediate, Low) against cumulative trait burden scores to demonstrate the compounding role of inheritance.
Diagnostic Proximity Band	Isolates borderline cases falling immediately adjacent	Provides a preventative care marker to classify	Donut Chart: Displays the breakdown of Clear

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	to structural boundaries (e.g., scores 4–5).	"Borderline/Monitor" individuals requiring active tracking.	Negative, Borderline/Monitor, and Clinical Range populations across the dataset.
Risk Overlap Profile	Categorizes individuals into four mutually exclusive biological brackets: Jaundice Only, Family ASD Only, Both, or Neither.	Examines the cumulative and compounding effects of early physiological and genetic vulnerabilities.	Grouped Bar Chart: Shows the actual diagnosis rate within each distinct biological intersection, validating that dual risk yields the highest statistical probability of ASD.
Generational Cohort	Bins age into standard sociological cohorts (Gen Alpha, Gen Z, Millennial, Gen X, Boomer).	Controls for historical screening biases and shifting clinical diagnostics over time.	Stacked Bar: Visualizes positive diagnostic proportions across cohorts, capturing historical variations in diagnostic criteria and awareness.

5. Statistical Testing & Validation Framework

To substantiate descriptive insights, a rigorous statistical framework evaluates differences across demographics and risk groups. These tests assess whether observed behavioral variations represent genuine population splits or random variation.

- **Chi-Square Tests of Independence:**

- *Application:* Applied across categorical distributions, specifically examining Gender × ASD Diagnosis, and Family History × Diagnosis status.
- *Purpose:* Verifies whether the higher diagnostic rate observed in females (31.7%) compared to males (28.2%) is statistically significant, validating genuine presentation skew rather than sampling variance. It similarly evaluates if biological risk factors are independently linked to ultimate outcomes.
- *Visual Support:* Validates the foundational variations observed in the **100% Stacked Bar Charts** and demographic cross-tabulated heatmaps.

- **Two-Sample T-Tests / Mann-Whitney U Tests:**

- *Application:* Utilized to compare continuous composite distributions (e.g., Total Atypical Trait Burden, Communication Deficit Score) across binary diagnostic outcomes (Positive vs. Negative).

- *Purpose:* Scientifically establishes if the continuous scoring systems successfully split the population. The test confirms that the mean behavioral load of the positive cohort is structurally distinct from the negative cohort.
- *Visual Support:* Provides the mathematical foundation backing the distinct separation seen in the **Bimodal Histograms** and **Box Plots**.
- **Analysis of Variance (ANOVA) / Kruskal-Wallis Tests:**
 - *Application:* Conducted across multi-level categorical features, including the four mutually exclusive bands of the *Risk Overlap Profile* and the multi-tiered *Behavioral Archetypes*.
 - *Purpose:* Validates that cumulative risk factors (Jaundice + Family History) produce statistically distinct behavioral burdens compared to isolated risks or baseline backgrounds. It also ensures that engineered archetypes possess statistically unique trait variances.
 - *Visual Support:* Grouped bar charts and 100% stacked phenotype distributions are backed by these multi-group variance verifications.